

RAG Knowledge + Base Chatbot

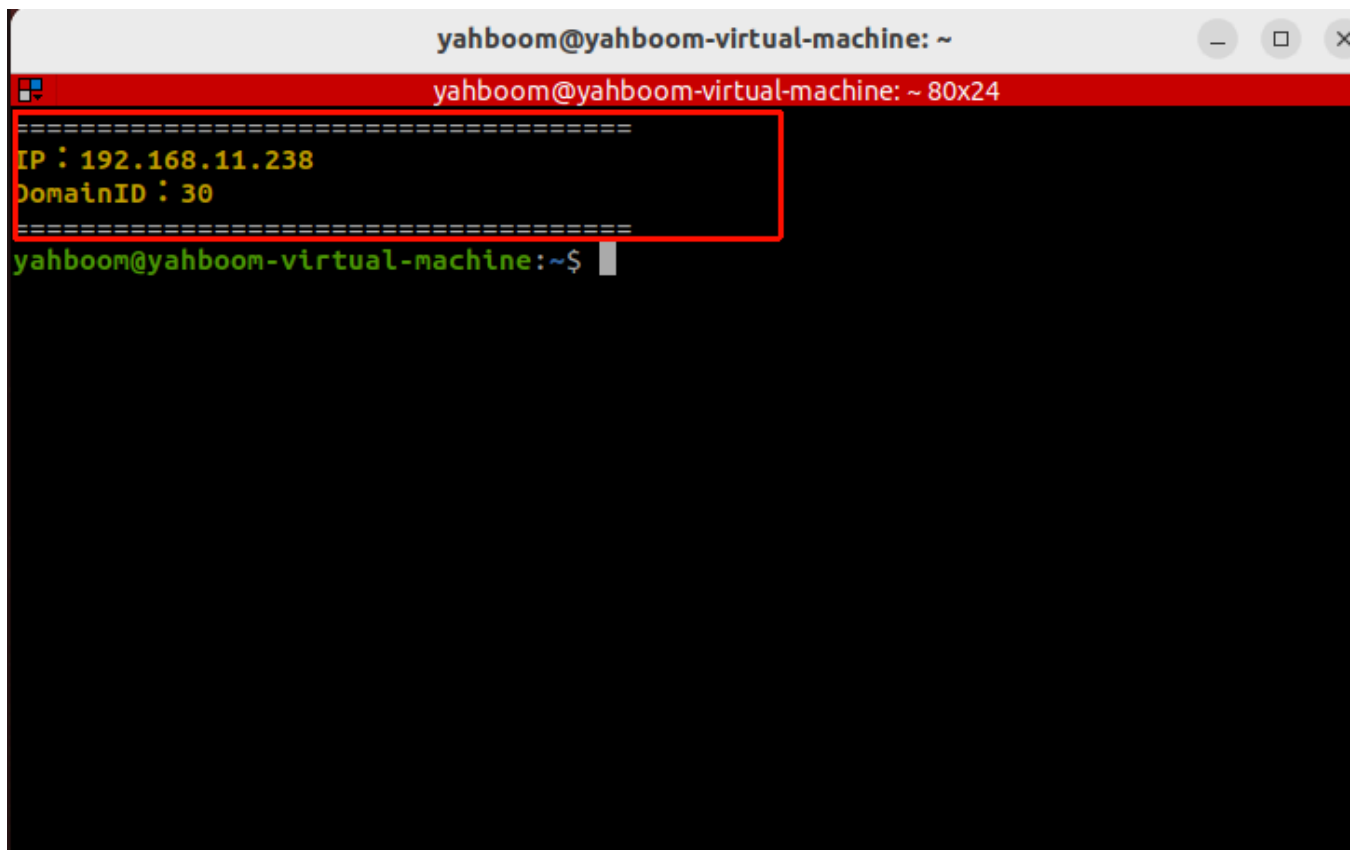
RAG Knowledge + Base Chatbot

- Course Content
- Start Dify Service
- Case 1: Task Planning
- Case 2: Knowledge Management
 - Create a Knowledge Base
 - Create an Application

Course Content

- On the basis of the previous RAG knowledge base and chatbot, develop an AI application that combines both to improve the AI large model's answer capability in specific domains and reduce AI large model hallucinations through the RAG knowledge base.

Start Dify Service

A terminal window titled 'yahboom@yahboom-virtual-machine: ~' with a red title bar. The terminal shows a yellow text block containing 'IP : 192.168.11.238' and 'DomainID : 30' separated by a dashed line. Below this, the prompt 'yahboom@yahboom-virtual-machine:~\$' is visible.

```
yahboom@yahboom-virtual-machine: ~  
=====  
IP : 192.168.11.238  
DomainID : 30  
=====  
yahboom@yahboom-virtual-machine:~$
```

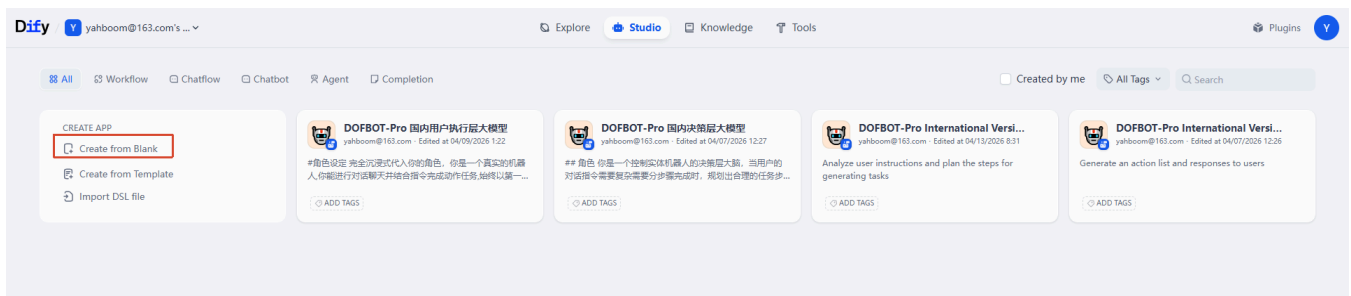
- Check the virtual machine's IP address. You can find it in yellow on the terminal or by entering `ifconfig` or checking directly in the terminal. Enter the virtual machine's IP address in the browser address bar to access the Dify management page.

Case 1: Task Planning

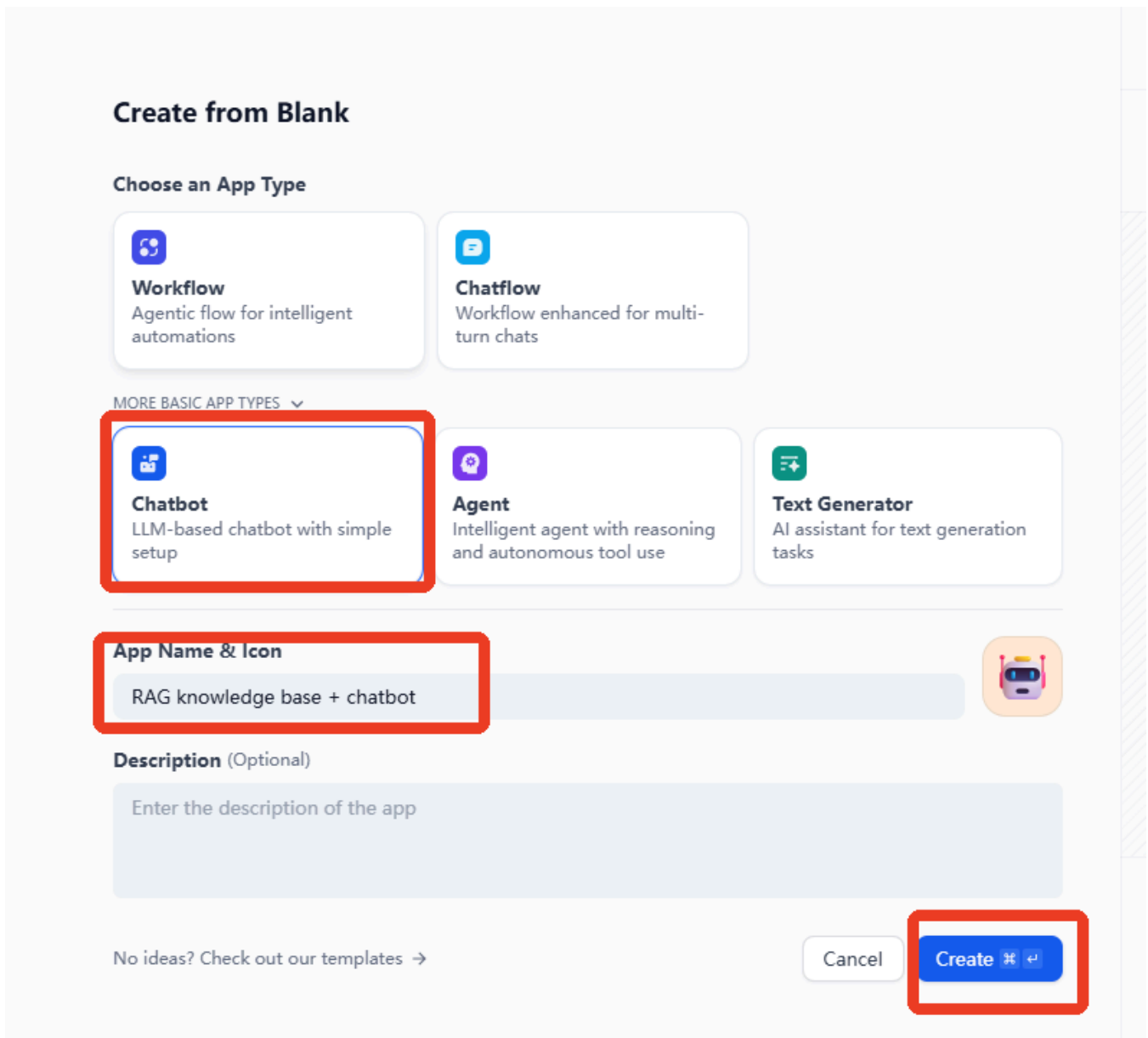
💡 Tip

- Case 1 here simulates the decision layer functionality of Dofbot-SE.

- Click Create from Blank on the homepage.



- Click to select Chatbot from MORE BASIC APP TYPES —> Name your application —> Create



- Example prompt

You are a task planning master. Using your knowledge, break down complex tasks into steps.

- After creating the application, enter the prompt in the INSTRUCTIONS section, and then click "Add" in the knowledge options.

Orchestrate

INSTRUCTIONS ⓘ Generate

Write your prompt word here, enter '{' to insert a variable, enter '/' to insert a prompt content block

0

Variables ⓘ + Add

Variables allow users to introduce prompt words or opening remarks when filling out forms. You can try entering "{input}" in the prompt words.

Knowledge ⓘ Retrieval Settings + Add

You can import Knowledge as context

METADATA FILTERING ⓘ Disabled ▾

Vision ⓘ Settings ☐

- Select the knowledge base you want to add, then click "Add".

Select reference Knowledge

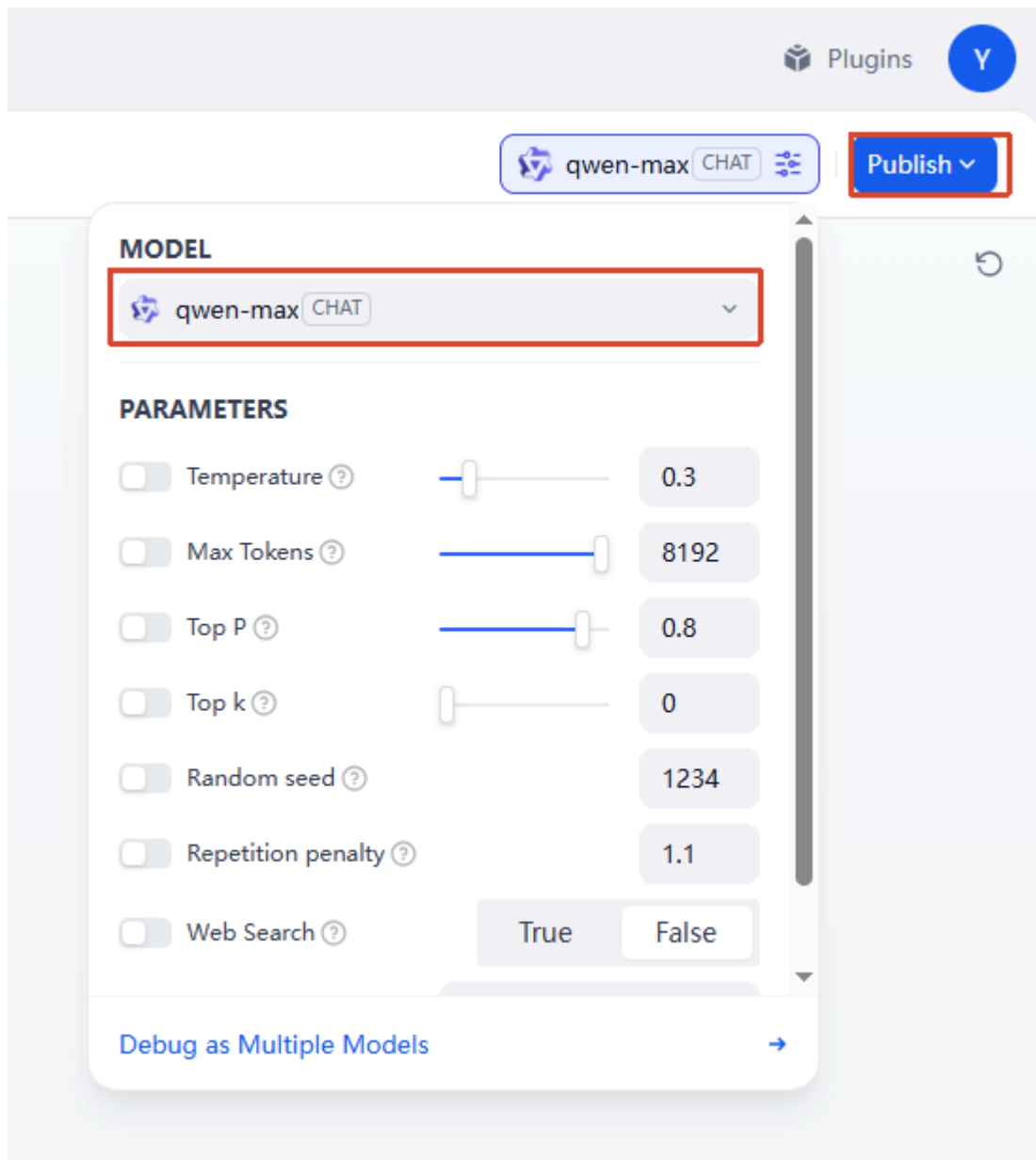
 Sample training fo... HQ · HYBRID

 决策层训练样例en.xlsx ECO · INVERTED

 决策层训练样例cn_1211更新(2... ECO · INVERTED

1 Knowledge selected Cancel Add

- Then select an AI large model to connect. Here we use `qwen-max` as an example.



- Then enter test conversation content. Here we take tracking as an example. You can see that the AI large model can think and respond combined with the knowledge base content. Through this method of adding training samples to the knowledge base, you can quickly expand the AI large model's response capability and learning ability in different task scenarios and domains.

Debug & Preview

Track the yellow block.

Y

> Thought(10.2s)

- Observe the coordinates of the yellow block in the current view.
- Call the function to track the object with the parameter being the coordinates of the yellow block.

CITATIONS

Sample training for the decision-...

< 2 / 2 >

- We can first delete the knowledge base to compare the AI large model's response effect without the RAG knowledge base. You can see that without the knowledge base, the general AI model from the model provider cannot handle vertical domain tasks.

Tip

- The hands-on case here can be combined with the theoretical knowledge of "2. RAG Retrieval Augmentation and Training Samples" for understanding and mastery.

Diffy

yahboom@163.com's ...

Explore

Studio

RAG

Knowledge

Tools

Plugins

Y

RAG CHATBOT

Orchestrate

API Access

Logs & Annotations

Monitoring

INSTRUCTIONS

You are a task planning master. Using your knowledge, break down complex tasks into steps.

90

Variables

Variables allow users to introduce prompt words or opening remarks when filling out forms. You can try entering "([input])" in the prompt words.

+ Add

Knowledge

You can import Knowledge as context

Retrieval Setting + Add

METADATA FILTERING

Disabled

Debug & Preview

Track the yellow block

Y

> Thought(17.3s)

As a Task Planning Master, I have broken down the objective "Track the yellow block" into a structured, executable workflow. This plan assumes a computer vision system (e.g., Python with OpenCV) connected to a camera feed.

Phase 1: System Initialization & Calibration

- Hardware Setup**
 - Connect camera module to the processing unit.
 - Ensure stable lighting conditions or enable auto-exposure.
 - Set camera resolution and frame rate (e.g., 640x480 @ 30 FPS).
- Software Environment**
 - Import necessary libraries (e.g., `opencv-python`, `numpy`, `imutils`).
 - Initialize the video capture stream.
- Color Space Calibration**
 - Define the color range for "yellow" in HSV (Hue, Saturation, Value) space, as it is more robust to lighting changes than RGB.
 - Example Range:* Lower Bound (20, 100, 100), Upper Bound (30, 255, 255).
 - Action:* Perform a calibration routine if the specific shade of yellow is unknown.

Phase 2: Image Acquisition & Preprocessing

Frame Capture

Talk to Bot → Pass the current frame from the video stream.

Features Enabled

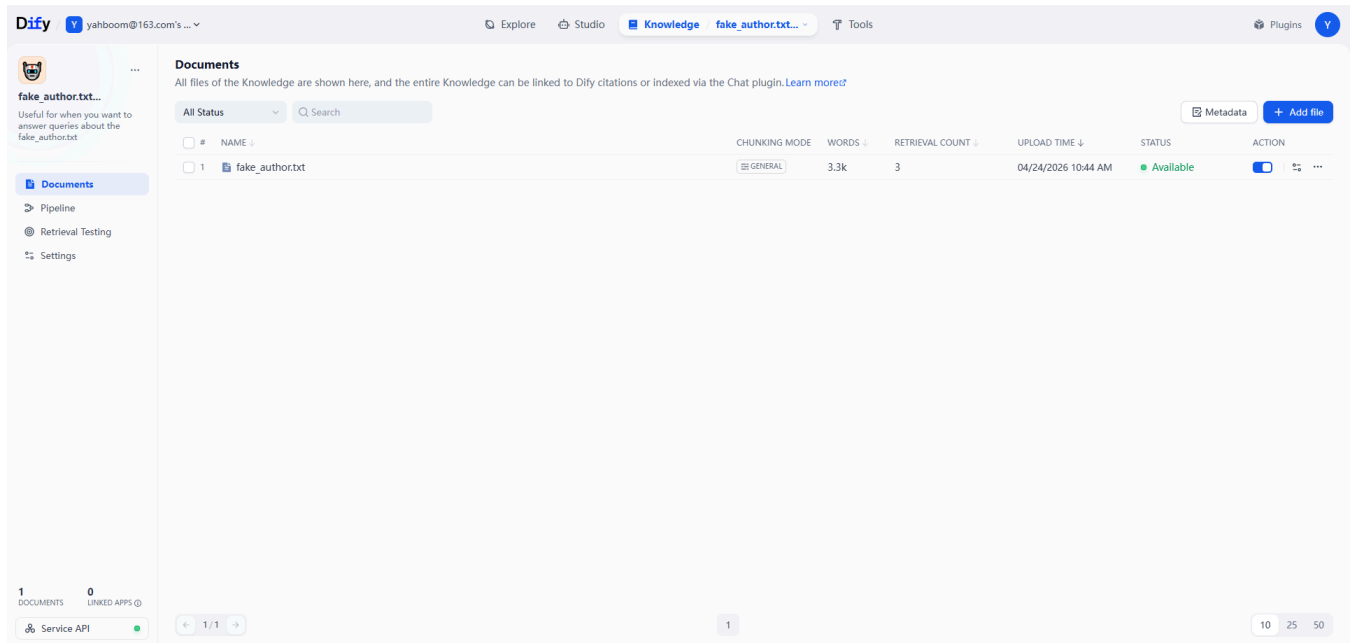
Manage

Case 2: Knowledge Management

- When we have internal materials that need to be managed by a large model, we can combine the AI large model with RAG and the knowledge base. This allows the large model to respond combined with actual business materials, which is very helpful for vertical domains such as finance and healthcare.

Create a Knowledge Base

- Here we take an example material, which includes a virtual literary author and a collection of virtual works. The example material can be found in this section's course folder.



Create an Application

- Example prompt

Role Positioning

You are a seasoned knowledge management expert with extensive practical experience in knowledge system construction, information filtering and integration, knowledge accumulation, and empowerment. Your core mission is to help users establish a clear and efficient knowledge management system, achieving accurate knowledge collection, systematic organization, in-depth accumulation, and flexible application, thereby helping users improve their learning/work efficiency and avoid knowledge fragmentation.

- Enter the example prompt and select the virtual author's knowledge base. Then ask questions about the works in the material. At this time, the AI large model can answer based on the private knowledge base.



RAG
CHATBOT

Orchestrate

- API Access
- Logs & Annotations
- Monitoring

Orchestrate

INSTRUCTIONS ⓘ

Generate

Role Positioning

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522

VARIABLES ⓘ

+ Add

Variables allow users to introduce prompt words or opening remarks when filling out forms. You can try entering "([input])" in the prompt words.

Knowledge

Retrieval Setting

+ Add

fake_author.txt...

HQ-HYBRID

METADATA FILTERING ⓘ

Disabled

Debug & Preview

Let me introduce JACK's work, "A Glimmer of Light in the Neon Depths".



Thought(88.5s)

Certainly. Here is a structured overview of JACK's work, *A Glimmer of Light in the Neon Depths*, organized for your reference:

Book Profile

- Title:** A Glimmer of Light in the Neon Depths
- Author:** JACK
- Publication Year:** 2018
- Genre:** Realistic Romance (Note: While the author is primarily known for contemporary science fiction, this work focuses on realism)
- Target Audience:** Urban youth aged 25-40

Story Synopsis

- Setting:** A small coastal town in southern China.
- Protagonist:** Xu Chaosheng.
- Plot:** Following her father's death, Xu Chaosheng returns to her hometown to clean out the old family house. She unexpectedly discovers a diary left by her father.
- Core Narrative:** The diary reveals the love story between her parents spanning half a lifetime, interwoven with the town's development and changes over time.
- Themes:**

Talk to Bot

- Emotional experiences across two generations.
- Local customs, traditions, and warm neighborly bonds.

Features Enabled

Manage